Azure AI Search vs SQL Server 2025 Hyperscale

Native vector + semantic search comparison · cost crossover model · migration map · East US pricing, May 2026

Your workload

AI SEARCH BASELINE TIER	PARTITIONS (TODAY)	REPLICAS (TODAY)	YOY DATA GROWTH %	TIER STORAGE UTILIZATION %
Standard S2 (512 GB/par	6	4	40	60

SEMANTIC RANKER QUERIES / MO (K)	SQL VCORES PER REPLICA	SQL HA REPLICAS (INCL. PRIMARY)	SQL COMPUTE TIER	SQL RESERVATION
500	32	4	Premium Mem-Opt ($0.2	1-year reserved (~35% o

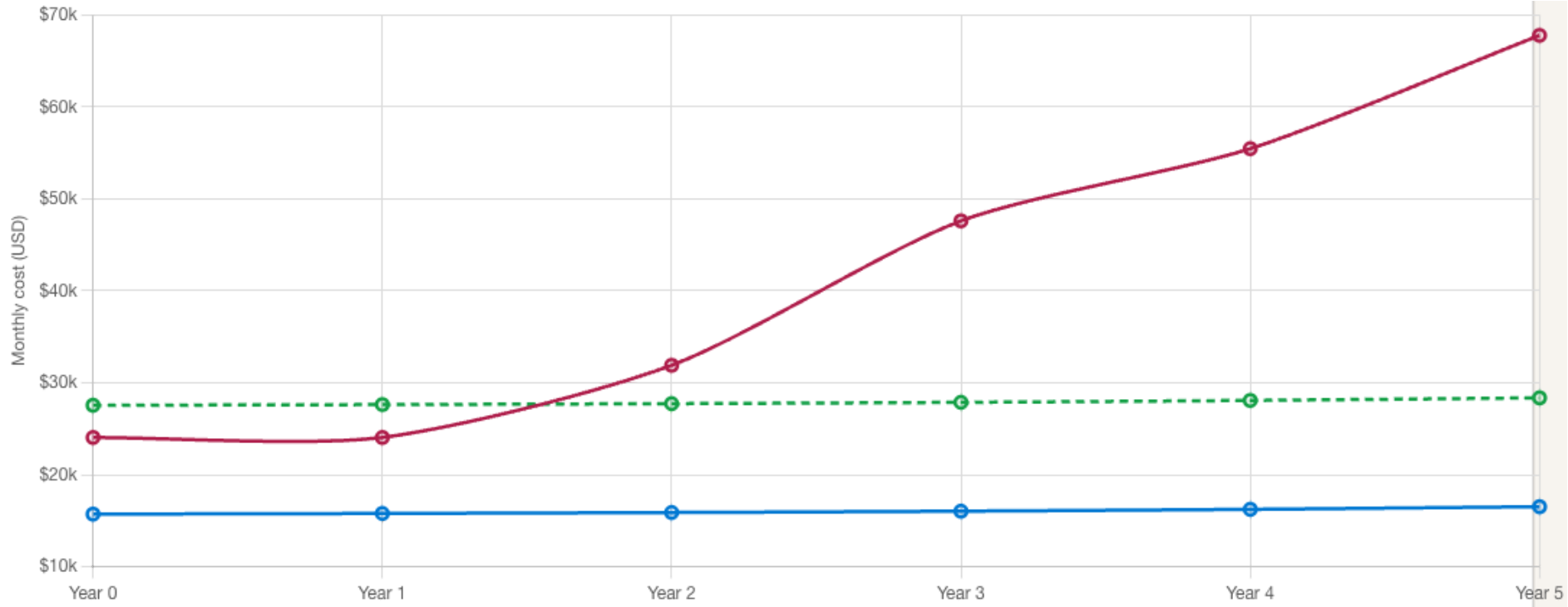
SQL OPS OVERHEAD $/MO (RECURRING)	SQL ONE-TIME BUILD COST $ (NRE)	PROJECTION HORIZON (YEARS)
6000	350000	5

COST CROSSOVER

~ Year 1.5

Through year 1.5, **Azure AI Search is cheaper**. After that, SQL Hyperscale + ops overhead crosses below as AI Search is forced into higher tiers.

5-year cost projection



▶ Assumptions and how the model works

Year-by-year breakdown

YEAR	DATA (TB)	AI SEARCH TIER	PARTITIONS × REPLICAS	AI SEARCH $/MO	SQL HS $/MO	SQL HS ALL-IN $/MO	CHEAPER
Year 0	1.80	Std S2	6 × 4 = 24 SU	$24,047	$15,716	$27,549	AI Search
Year 1	2.52	Std S2	6 × 4 = 24 SU	$24,047	$15,789	$27,623	AI Search
Year 2	3.53	Std S2	8 × 4 = 32 SU	$31,896	$15,893	$27,726	SQL Hyperscale
Year 3	4.94	Std S3	6 × 4 = 24 SU	$47,594	$16,037	$27,871	SQL Hyperscale

Year 4	6.91	Std S3	7 × 4 = 28 SU	$55,443	$16,239	$28,073	SQL Hyperscale
Year 5	9.68	Storage Opt L1	6 × 4 = 24 SU	$67,759	$16,523	$28,356	SQL Hyperscale

Capability parity matrix

CAPABILITY	AZURE AI SEARCH	SQL SERVER 2025 HYPERSCALE	PARITY
Vector type & distance	HNSW + exhaustive KNN, cosine/dot/Euclidean, up to **4 096 dims**, scalar & binary quantization, narrow types (int8, float16)	`vector(N)` data type up to **1 998 dims**, `VECTOR_DISTANCE` function with cosine/dot/Euclidean. *Blocks `text-embedding-3-large` at 3 072 dims.*	Partial
ANN vector index	HNSW, managed automatically per index, GA, vector compression baked in	`CREATE VECTOR INDEX … USING DiskANN` — **still in public preview** as of May 2026. Requires `PREVIEW_FEATURES = ON`. Exact KNN only for < 50K vectors.	Preview
Lexical / BM25 search	Built-in BM25 with analyzers, synonyms, stop-words, language packs	Full-Text Search (`CONTAINS`/`FREETEXT`) + Semantic Search (key phrases / similar docs)	Yes
Hybrid (vector + lexical)	Native Reciprocal Rank Fusion (RRF), single query, single response	Hand-rolled CTE combining FTS rank + `VECTOR_DISTANCE` with manual RRF	DIY
Semantic re-ranker (L2)	Microsoft-hosted cross-encoder, billed per query, captions + answers + query rewrite	Not built-in. Requires self-hosted cross-encoder or AOAI re-rank call	Gap
Integrated vectorization	One-click chunk + embed at index time via AOAI / AI Foundry	`sp_invoke_external_rest_endpoint` to AOAI, or external pipeline	DIY
Indexers / ingestion connectors	Blob, ADLS, Cosmos, SQL DB, Fabric OneLake, SharePoint, etc. — managed change tracking	ADF, Fabric pipelines, Logic Apps, or custom Functions — manual change tracking	Gap
Document cracking (PDF/DOCX/HTML)	Document Extraction skill + OCR + custom skillsets	Azure AI Document Intelligence called externally; raw text stored in SQL	External

Faceting & filtering	Native facets, filterable fields, scoring profiles	`GROUP BY`, `WHERE`, computed/indexed columns, materialized views	Yes
Suggesters / autocomplete	Built-in suggester data structure with fuzzy matching	Trigram or prefix indexes, hand-built	DIY
Geo search	Native `Edm.GeographyPoint`, geo distance & polygon filters	Native `geography` / `geometry` spatial types, spatial indexes	Yes
Agentic retrieval	Native query planner, conversation context, BYOM, billed per reasoning token	Not built-in; orchestrate from app tier with Semantic Kernel / LangChain	Gap
HA / read scale-out	Replicas (1–12), automatic load-balancing	Hyperscale HA replicas (1–4) + named replicas for read scale	Yes
Vertical scale (CPU/RAM)	Fixed per tier — must upgrade tier (S1→S2→S3)	Independent vCore and storage scaling, online vCore changes	Better
Transactional writes	Eventually consistent index updates; no ACID across documents	Full ACID, joins, transactions, foreign keys — same engine as your OLTP	Better
Security: CMK, private link, RBAC	CMK, private endpoints, Entra ID auth, role-based access	TDE w/ CMK, private link, Entra auth, RLS, column encryption	Yes
Backup / PITR	No user-controlled backups (re-index from source)	Continuous PITR, 1–35 day retention, LTR up to 10 years	Better
SLA	99.9% read for ≥2 replicas, 99.9% read/write for ≥3 replicas	99.99% for ≥1 HA replica, 99.995% for zone-redundant HA	Better

Sources: Azure AI Search service limits (link), Azure SQL Hyperscale docs (link), SQL Server 2025 vectors (link).

Can SQL Server 2025 replace Azure AI Search with zero effort?

No. SQL Server 2025 Hyperscale gives you the *primitives* for a retrieval system — the `vector` data type, DiskANN (preview), Full-Text Search, statistical semantic key phrases. Azure AI Search is a *turnkey retrieval platform*. To get from primitives to platform there is meaningful upfront engineering, ongoing operational overhead, and a small set of capabilities that don't have a SQL equivalent at all.

What does NOT port at all (no SQL equivalent — must approximate or live without)

AI SEARCH FEATURE	CLOSEST SQL SUBSTITUTE	PARITY
Microsoft-tuned semantic re-ranker (L2)	Self-host a cross-encoder (e.g. ms-marco-MiniLM) on Container Apps + GPU, or call AOAI gpt-4o-mini for LLM re-rank. Quality and per-query latency will differ.	Approximate only
Agentic retrieval (autonomous query planner)	Orchestrate from app tier with Semantic Kernel or LangGraph against your SQL hybrid search. Built, not bought.	Build yourself
Integrated vectorization (one-click chunk + embed at index time)	External pipeline calling AOAI embeddings, or `sp_invoke_external_rest_endpoint` per row. No UI, no managed orchestration.	Build yourself
Indexer connector library (Blob, ADLS, Cosmos, SQL DB, Fabric, SharePoint, ServiceNow…)	ADF / Fabric pipelines / custom Functions. You build and operate each connector and its change-tracking.	Build yourself
4 096-dim vector fields (supports text-embedding-3-large at 3 072)	SQL Server 2025 hard-caps at **1 998 dims**. You must use a smaller embedding model (ada-002 at 1 536 fits; 3-large at 3 072 does not).	Hard ceiling
Production-grade ANN vector index (GA)	DiskANN is **public preview** in May 2026. Exact KNN works but only practical under ~50 K vectors.	Preview risk
Native suggesters / autocomplete with fuzzy match	Trigram or prefix index, hand-built ranker.	DIY

Upfront engineering build-out (one-time, NRE)

WORKSTREAM	ENG-WEEKS (RANGE)	NOTES

Schema, Hyperscale provisioning, networking, identity	1–2	Straightforward.
Ingestion pipeline (replace indexers)	4–8	One pipeline per source system; change tracking.
Document cracking (Document Intelligence integration)	1–2	If you have rich PDF/DOCX content.
Chunking pipeline	1–2	Recursive splitter or domain-aware.
Embedding generation pipeline	1–2	Batch via Functions / Fabric. Avoid per-row `sp_invoke_external_rest_endpoint`.
DiskANN vector index setup + tuning	1–2	Preview feature; expect iteration.
Hybrid retrieval w/ RRF in T-SQL	2–4	One stored procedure per query shape; hand-coded RRF.
Semantic re-ranker (host or call AOAI)	4–8	Biggest individual line item. GPU container app + model serving + latency tuning.
Faceting, filters, scoring profiles	2–4	Per query/result shape.
Observability & relevance dashboards	2–3	You lose AI Search query telemetry; rebuild it.
Testing, NDCG/MRR benchmarking, cutover	4–8	Shadow traffic, A/B, parity sign-off.
Total	**23–45 eng-weeks**	**≈ 2 senior engineers for 12–22 calendar weeks.**

At a fully-loaded $8 K / engineer-week (US senior IC, blended), the build cost lands roughly **$184 K – $720 K one-time**. The model's default is $350 K. Add the cost-tab "SQL one-time build cost" input to fold it into the projection as straight-line amortization across your horizon.

15-step build playbook

In order. Where parity is lost, called out inline.

1 **Stand up the database**

Provision Azure SQL Database Hyperscale (Standard Gen5 or Premium Mem-Opt). Enable Premium-series memory-optimized compute for vector workloads. Configure 1–4 HA replicas to mirror your AI Search replica count and one or more named replicas for read scale-out.

2 **Design the schema**

Create a `documents` table for source metadata and a `chunks` table with `chunk_id`, `doc_id`, `text nvarchar(max)`, `embedding vector(N)`, plus filterable / facetable columns. Add full-text catalog on the text column.

3 **Replace indexers — build an ingestion pipeline**

AI Search indexers (Blob, ADLS, Cosmos, SQL, Fabric, SharePoint) become Azure Data Factory pipelines, Fabric Dataflows, or Azure Functions. You own change tracking (high-water mark column or CDC).

4 **Replace document cracking**

Call Azure AI Document Intelligence for PDF/Office/OCR. Land raw extracted text in `documents.raw_text`. AI Search did this inline; here it's an external skill in the pipeline.

5 **Replace the Text Split skill — chunking**

Implement chunking in the pipeline (LangChain RecursiveCharacterTextSplitter, Semantic Kernel, or your own UDF). Persist chunks as rows in `chunks`.

6 **Replace integrated vectorization — embeddings**

Generate embeddings with Azure OpenAI `text-embedding-3-large`. Either call `sp_invoke_external_rest_endpoint` from a stored procedure (per-row, simple) or batch externally in the pipeline (faster, cheaper). Store as `vector(1536)` or `vector(3072)` — note SQL Server 2025 caps at 1 998 dimensions.

7 **Build the vector index**

`CREATE VECTOR INDEX IX_chunks_emb ON dbo.chunks(embedding) WITH (METRIC = 'cosine', TYPE = 'diskann');` Note: at GA the DiskANN index is single-replica and rebuilt from scratch — plan for batched maintenance windows.

8 **Re-implement hybrid retrieval with RRF**

Write a single query that runs an FTS lexical search, a vector KNN, ranks each, then combines with hand-coded Reciprocal Rank Fusion. AI Search does this in one line; here it's ~40 lines of CTE/T-SQL per query shape.

9 Build a semantic re-ranker — the biggest gap

Choose one: (a) self-host a cross-encoder (e.g. `ms-marco-MiniLM-L-12-v2`) on Azure Container Apps or AKS with GPU and call it from the app tier on the top-50 results; (b) call Azure OpenAI `gpt-4o-mini` to score relevance — much higher quality, much higher cost. Either way: no built-in equivalent to AI Search semantic ranker.

10 Recreate scoring profiles and synonyms

Move field-weighting, freshness boosts, and tag boosts into your T-SQL ranking expression. Use FTS thesaurus XML for synonyms; stop-lists for stop words.

11 Recreate facets, suggesters, and autocomplete

Facets → `GROUP BY` with indexed columns or pre-aggregated materialized views. Suggesters → prefix or trigram index on the suggester source field; query with `LIKE 'term%'`.

12 Wire up observability & query telemetry

AI Search Insights → use Azure SQL Database Insights, Query Performance Insight, and Application Insights from the app tier. Build a custom relevance dashboard since AI Search's per-query telemetry is not available.

13 Security, network, and identity

Enable TDE with customer-managed key, private endpoints, Entra-only auth, row-level security if you have multi-tenant data, and column-level encryption for PII. All native in Hyperscale.

14 If you used agentic retrieval — rebuild it

AI Search's agentic retrieval (query planning + reasoning tokens) has no SQL equivalent. Build it in the app tier with Semantic Kernel or LangGraph orchestrating multi-turn retrieval against your SQL hybrid search.

15 Cut over with shadow traffic

Run both services in parallel. Replay a representative query log against both and compare top-K results with NDCG / MRR. Only cut over when SQL hybrid + custom re-rank beats AI Search baseline on your relevance benchmark.

Biggest gaps when moving off AI Search: the semantic re-ranker, integrated vectorization, indexer-style connectors, and agentic retrieval. These are

real engineering work — budget them into the cost model on the previous tab via the "SQL ops overhead" input.

Production readiness flag (May 2026): The DiskANN *vector index* in SQL Server 2025 / Azure SQL is still **public preview** — not GA. Exact KNN (no index) is GA but only practical for < ~50K vectors. If your workload needs ANN performance on millions of vectors today, AI Search HNSW is the only GA path on Microsoft's stack. Revisit when DiskANN GAs.

Build playbook (detailed) — execute each step end-to-end

Below is the operator-grade walkthrough for the 15-step build. Every step shows: **goal**, **prerequisites**, **commands / code**, **sample execution**, **validation check**, and the **common pitfalls**. All Azure CLI samples assume `az` ≥ 2.60. SQL samples assume Azure SQL Database Hyperscale (the SQL Server 2025 vector engine). Python samples assume Python 3.11 with `azure-identity`, `azure-ai-documentintelligence`, `openai`, `pyodbc`.

Variables used throughout: `RG=rg-search-prod` · `LOC=eastus` · `SRV=sql-search-prod` · `DB=searchdb` · `AOAI=aoai-search-prod` · `DOCAI=docai-search-prod`.

15 STEPS

01 Stand up the database

Goal: a production Hyperscale instance with HA + named-replica read scale-out and the preview switch flipped so DiskANN works.

PREREQUISITES

- Azure subscription with quota for *Premium Mem-Optimized Gen5* in your region
- VNet + subnet for the private endpoint
- An Entra group to act as the SQL admin (no SQL auth)

EXECUTE

PROVISION.SH **bash**

```bash
RG=rg-search-prod
LOC=eastus
SRV=sql-search-prod
DB=searchdb
ADMIN_GROUP="grp-sql-search-admins"
ADMIN_GROUP_OID=$(az ad group show --group "$ADMIN_GROUP" --query id -o tsv)

az group create -n $RG -l $LOC

# Server with Entra-only auth
az sql server create -g $RG -n $SRV -l $LOC \
  --enable-ad-only-auth \
  --external-admin-principal-type Group \
  --external-admin-name "$ADMIN_GROUP" \
  --external-admin-sid $ADMIN_GROUP_OID

# Hyperscale database, Premium Mem-Opt for vector workloads
az sql db create -g $RG -s $SRV -n $DB \
  --edition Hyperscale \
  --family Gen5 \
  --compute-model Provisioned \
  --tier Hyperscale \
  --capacity 16 \
  --read-scale Enabled \
  --high-availability-replica-count 1 \
  --backup-storage-redundancy Zone \
  --zone-redundant true

# Optional: one named replica for read scale-out (mirrors AI Search replica idea)
az sql db replica create -g $RG -s $SRV -n $DB \
  --partner-server $SRV --partner-database ${DB}-ro \
  --secondary-type Named --capacity 8 --family Gen5
```

FLIP THE PREVIEW SWITCH (REQUIRED FOR DISKANN, MAY 2026)

ENABLE-PREVIEW.SQL sql

```sql
-- Run as a member of the Entra admin group
ALTER DATABASE SCOPED CONFIGURATION SET PREVIEW_FEATURES = ON;

SELECT name, value, is_value_default
FROM sys.database_scoped_configurations
WHERE name = 'PREVIEW_FEATURES';
```

SAMPLE OUTPUT

```
name              value   is_value_default
PREVIEW_FEATURES  1       0
```

Validation: `SELECT @@VERSION;` returns *Microsoft SQL Azure (RTM) – 16.x* with the 2025 build; `SELECT SERVERPROPERTY('EngineEdition')` = 5 (Hyperscale).

Pitfall: picking Standard Gen5 vCore (not Premium Mem-Opt) means vector queries fall back to row-mode execution and DiskANN training is >5× slower. Spend the extra ~40% on Premium for any meaningful vector workload.

02 Design the schema

Goal: a normalized `documents` + `chunks` shape that supports vector search, full-text search, faceting, and audit — equivalent to an AI Search index with sub-document chunks.

EXECUTE

SCHEMA.SQL sql

```sql
CREATE SCHEMA search;
GO

CREATE TABLE search.documents (
  doc_id          UNIQUEIDENTIFIER NOT NULL DEFAULT NEWID() PRIMARY KEY,
  source_uri      NVARCHAR(1024)   NOT NULL,
  source_etag     NVARCHAR(128)    NULL,
  title           NVARCHAR(512)    NULL,
  author          NVARCHAR(256)    NULL,
  tags            NVARCHAR(MAX)    NULL,          -- JSON array
```

```
  language        CHAR(2)          NULL,
  modified_at     DATETIME2(3)     NOT NULL,
  raw_text        NVARCHAR(MAX)    NULL,
  ingested_at     DATETIME2(3)     NOT NULL DEFAULT SYSUTCDATETIME(),
  CONSTRAINT UQ_documents_source UNIQUE (source_uri)
);

CREATE INDEX IX_documents_modified ON search.documents(modified_at);
CREATE INDEX IX_documents_author   ON search.documents(author);

CREATE TABLE search.chunks (
  chunk_id        BIGINT IDENTITY(1,1) PRIMARY KEY,
  doc_id          UNIQUEIDENTIFIER NOT NULL,
  ordinal         INT NOT NULL,
  text            NVARCHAR(MAX) NOT NULL,
  embedding       VECTOR(1536) NULL,              -- text-embedding-3-small
  embedding_model NVARCHAR(64) NULL,
  token_count     INT NULL,
  created_at      DATETIME2(3) NOT NULL DEFAULT SYSUTCDATETIME(),
  CONSTRAINT FK_chunks_doc FOREIGN KEY (doc_id)
    REFERENCES search.documents(doc_id) ON DELETE CASCADE
);

CREATE INDEX IX_chunks_doc ON search.chunks(doc_id, ordinal);

-- Full-text catalog for lexical / BM25-like search
CREATE FULLTEXT CATALOG ft_search AS DEFAULT;
CREATE FULLTEXT INDEX ON search.chunks(text LANGUAGE 1033)
  KEY INDEX PK__chunks__... ON ft_search WITH CHANGE_TRACKING AUTO;
```

Validation:

```
SELECT t.name, c.name, ty.name AS dtype, c.max_length
FROM sys.tables t JOIN sys.columns c ON c.object_id = t.object_id
JOIN sys.types ty ON ty.user_type_id = c.user_type_id
WHERE t.name IN ('documents','chunks') ORDER BY t.name, c.column_id;
```

Expect the `embedding` column to show type `vector` with `max_length = 6148` (1536 dims × 4 bytes + 4-byte header).

Pitfall: `VECTOR(N)` in SQL Server 2025 caps at **1 998 dimensions**. `text-embedding-3-large` (3072 dims) *does not fit*. Use `text-embedding-3-small` (1536 dims) or truncate large embeddings with the AOAI `dimensions` parameter to stay under the cap.

03 Replace indexers — build an ingestion pipeline

Goal: reliably pick up new / changed source documents and land them in `search.documents`, then trigger chunk + embed downstream. AI Search indexers are a managed scheduler with built-in change tracking — here you own that.

REFERENCE ARCHITECTURE

- **Trigger:** Azure Data Factory tumbling-window trigger every 15 min, or Event Grid for instant ingestion from Blob.
- **Pattern:** high-water mark (`modified_at > LAST_RUN_TS`) for SQL/CDC sources; ETag/ `LastModified` headers for Blob/ADLS; `_ts` filter for Cosmos DB.
- **State:** store last-run-cursor in a small bookkeeping table `search.ingestion_state`.

FUNCTION_APP.PY **python**

```python
import os, hashlib, logging
import azure.functions as func
from azure.identity import DefaultAzureCredential
from azure.storage.blob import BlobServiceClient
import pyodbc

app = func.FunctionApp()
SQL_CONN = os.environ["SQL_CONN"]   # uses Azure AD interactive / MSI
BLOB_URL = os.environ["BLOB_URL"]

@app.schedule(schedule="0 */15 * * * *", arg_name="timer")
def ingest(timer: func.TimerRequest) -> None:
    cred = DefaultAzureCredential()
    svc = BlobServiceClient(account_url=BLOB_URL, credential=cred)
    container = svc.get_container_client("corpus")

    with pyodbc.connect(SQL_CONN, autocommit=False) as cn:
        cur = cn.cursor()
        cur.execute("SELECT last_run FROM search.ingestion_state WHERE name='blob-corpus'")
        last_run = cur.fetchone()[0]

        upserts = 0
        for blob in container.list_blobs():
```

```
            if blob.last_modified <= last_run:
                continue
            etag = blob.etag.strip('"')
            cur.execute("""
                MERGE search.documents AS t
                USING (SELECT ? AS source_uri, ? AS source_etag,
                              ? AS title, ? AS modified_at) AS s
                ON t.source_uri = s.source_uri
                WHEN MATCHED AND t.source_etag <> s.source_etag THEN
                  UPDATE SET source_etag = s.source_etag,
                             modified_at = s.modified_at,
                             raw_text = NULL  -- force re-crack
                WHEN NOT MATCHED THEN
                  INSERT (source_uri, source_etag, title, modified_at)
                  VALUES (s.source_uri, s.source_etag, s.title, s.modified_at);
            """, f"{BLOB_URL}/corpus/{blob.name}", etag, blob.name, blob.last_modified)
            upserts += 1

        cur.execute("UPDATE search.ingestion_state SET last_run = SYSUTCDATETIME() WHERE name='blob-corpus'")
        cn.commit()
        logging.info(f"Ingest done: {upserts} upserts")
```

SAMPLE EXECUTION

```
2026-05-21T02:00:01Z  [INFO]  Ingest done: 137 upserts
2026-05-21T02:15:01Z  [INFO]  Ingest done: 12 upserts
2026-05-21T02:30:01Z  [INFO]  Ingest done: 0 upserts
```

Validation: after one run, `SELECT COUNT(*) FROM search.documents WHERE raw_text IS NULL;` shows the queue for Step 4 (cracking).

Pitfall: resetting the high-water mark to "0" silently re-ingests everything and re-embeds (Step 6) at full cost. Always re-ingest by deleting specific rows from `ingestion_state` AND zeroing `raw_text` for the affected docs only.

04 Replace document cracking

Goal: extract text from PDFs, Office docs, scanned images. Equivalent of the built-in cracking skill in AI Search.

PROVISION DOCUMENT INTELLIGENCE

DOCAI.SH **bash**

```bash
az cognitiveservices account create \
  -g $RG -n $DOCAI -l $LOC \
  --kind FormRecognizer --sku S0 \
  --custom-domain $DOCAI \
  --assign-identity
```

CRACKER FUNCTION (DRAINS `RAW_TEXT IS NULL` QUEUE)

CRACK.PY **python**

```python
from azure.ai.documentintelligence import DocumentIntelligenceClient
from azure.identity import DefaultAzureCredential
import pyodbc, os

cred = DefaultAzureCredential()
di = DocumentIntelligenceClient(
    endpoint=os.environ["DOCAI_ENDPOINT"], credential=cred)

cn = pyodbc.connect(os.environ["SQL_CONN"], autocommit=False)
cur = cn.cursor()
cur.execute("""
  SELECT TOP 50 doc_id, source_uri
  FROM search.documents
  WHERE raw_text IS NULL
  ORDER BY modified_at
""")
rows = cur.fetchall()

for doc_id, uri in rows:
    poller = di.begin_analyze_document(
        "prebuilt-layout", AnalyzeDocumentRequest(url_source=uri))
    result = poller.result()
    text = "\n\n".join(p.content for p in result.paragraphs)
    cur.execute("UPDATE search.documents SET raw_text = ? WHERE doc_id = ?",
                text, doc_id)
cn.commit()
print(f"Cracked {len(rows)} docs")
```

SAMPLE EXECUTION

```
$ python crack.py
Cracked 50 docs
$ python crack.py
Cracked 50 docs
$ python crack.py
Cracked 37 docs
```

Pitfall: `prebuilt-layout` costs ~$10/1K pages. For pure-text PDFs use `prebuilt-read` (~$1.50/1K). Profile your corpus first — running layout on a 1M-page corpus = $10K just to crack.

05 Replace the Text Split skill — chunking

Goal: split each document's `raw_text` into overlapping chunks that fit your embedding model's token window, persisted as rows in `search.chunks`.

CHUNK.PY — python

```python
from langchain.text_splitter import RecursiveCharacterTextSplitter
import pyodbc, tiktoken, os

splitter = RecursiveCharacterTextSplitter(
    chunk_size=800,           # tokens
    chunk_overlap=120,
    length_function=lambda s: len(tiktoken.get_encoding("cl100k_base").encode(s)),
    separators=["\n\n", "\n", ". ", " ", ""])

cn = pyodbc.connect(os.environ["SQL_CONN"], autocommit=False)
cur = cn.cursor()

# Docs that have text but no chunks yet
cur.execute("""
  SELECT d.doc_id, d.raw_text
  FROM search.documents d
  WHERE d.raw_text IS NOT NULL
    AND NOT EXISTS (SELECT 1 FROM search.chunks c WHERE c.doc_id = d.doc_id)
""")
for doc_id, text in cur.fetchall():
    for i, chunk in enumerate(splitter.split_text(text)):
        cur.execute("""
          INSERT INTO search.chunks (doc_id, ordinal, text, token_count)
          VALUES (?, ?, ?, ?)""",
```

```
            doc_id, i, chunk,
            len(tiktoken.get_encoding("cl100k_base").encode(chunk)))
cn.commit()
```

SAMPLE ROW AFTER RUN

```
chunk_id  doc_id        ordinal  token_count  text
--------  ------------  -------  -----------  ---------------------------------------
1         a2f9...c1     0        742          "Members enrolled in HMO plans must..."
2         a2f9...c1     1        788          "...selection of primary care provider..."
3         a2f9...c1     2        611          "...annual deductibles apply to..."
```

Pitfall: chunking on raw character count instead of tokens means ~30% of your chunks will exceed the embedding model's 8 191 token window and silently truncate. Always tokenize.

06 Replace integrated vectorization — embeddings

Goal: embed every `chunks.text` with Azure OpenAI and store the vector. AI Search calls AOAI for you; here you own batching, retries, throttling.

OPTION A — EXTERNAL BATCHER (PREFERRED FOR THROUGHPUT)

EMBED.PY — python

```python
from openai import AzureOpenAI
from azure.identity import DefaultAzureCredential, get_bearer_token_provider
import pyodbc, os, struct

cred = DefaultAzureCredential()
token = get_bearer_token_provider(cred, "https://cognitiveservices.azure.com/.default")
client = AzureOpenAI(
    azure_endpoint=os.environ["AOAI_ENDPOINT"],
    azure_ad_token_provider=token,
    api_version="2024-10-21")
MODEL = "text-embedding-3-small"   # 1536 dims, fits in VECTOR(1536)

cn = pyodbc.connect(os.environ["SQL_CONN"], autocommit=False)
cur = cn.cursor()

BATCH = 64
while True:
```

```
    cur.execute("""
      SELECT TOP (?) chunk_id, text
      FROM search.chunks WHERE embedding IS NULL ORDER BY chunk_id
    """, BATCH)
    rows = cur.fetchall()
    if not rows: break

    resp = client.embeddings.create(model=MODEL, input=[r.text for r in rows])
    for r, e in zip(rows, resp.data):
        # SQL Server accepts vector literal as JSON array string
        vec_json = "[" + ",".join(f"{v:.6f}" for v in e.embedding) + "]"
        cur.execute("""
          UPDATE search.chunks
          SET embedding = CAST(? AS VECTOR(1536)),
              embedding_model = ?
          WHERE chunk_id = ?
        """, vec_json, MODEL, r.chunk_id)
    cn.commit()
    print(f"Embedded batch of {len(rows)}")
```

OPTION B — IN-DATABASE VIA `SP_INVOKE_EXTERNAL_REST_ENDPOINT`

EMBED_IN_DB.SQL **sql**

```
-- One-time: managed identity + scoped credential to AOAI
CREATE DATABASE SCOPED CREDENTIAL AOAICred
WITH IDENTITY = 'Managed Identity';

CREATE OR ALTER PROCEDURE search.embed_chunk @chunk_id BIGINT
AS
BEGIN
  DECLARE @text NVARCHAR(MAX), @payload NVARCHAR(MAX), @resp NVARCHAR(MAX);
  SELECT @text = text FROM search.chunks WHERE chunk_id = @chunk_id;

  SET @payload = JSON_OBJECT('input': @text, 'model': 'text-embedding-3-small');

  EXEC sp_invoke_external_rest_endpoint
    @url      = 'https://aoai-search-prod.openai.azure.com/openai/deployments/te3s/embeddings?api-version=2024-10-21',
    @method   = 'POST',
    @payload  = @payload,
    @credential = AOAICred,
    @response = @resp OUTPUT;
```

```
  DECLARE @vec NVARCHAR(MAX) =
    (SELECT value FROM OPENJSON(@resp, '$.result.data[0].embedding'));

  UPDATE search.chunks
    SET embedding = CAST(@vec AS VECTOR(1536)),
        embedding_model = 'text-embedding-3-small'
    WHERE chunk_id = @chunk_id;
END
```

SAMPLE EXECUTION

```
$ python embed.py
Embedded batch of 64
Embedded batch of 64
Embedded batch of 64
...
Embedded batch of 17
```

Validation: `SELECT VECTOR_DIMS(embedding) FROM search.chunks WHERE chunk_id = 1;` returns **1536**.

Pitfall A: Option B (per-row stored proc) hits AOAI request-per-minute quotas FAST at scale. Use it only for trickle re-embeds; do bulk in Option A.

Pitfall B: if you re-embed with a new model, set `embedding_model` too and consider re-indexing — DiskANN's centroids are model-specific.

07 Build the vector index

Goal: create the DiskANN ANN index so KNN search is sub-second on millions of vectors. Without it, you're doing brute-force scans.

VECTOR-INDEX.SQL — sql

```
-- Confirm preview features are on (Step 1)
SELECT value FROM sys.database_scoped_configurations WHERE name = 'PREVIEW_FEATURES';

CREATE VECTOR INDEX IX_chunks_emb_diskann
ON search.chunks(embedding)
WITH (
  METRIC = 'cosine',
  TYPE   = 'diskann',
  MAXDOP = 8
```

```
);

-- Track build progress (build is async on Hyperscale)
SELECT name, type_desc, is_disabled, has_filter
FROM sys.indexes WHERE object_id = OBJECT_ID('search.chunks');

SELECT * FROM sys.dm_db_xtp_hash_index_stats; -- runtime stats
```

SAMPLE EXECUTION TIMELINE

```
2026-05-21 03:00:00  CREATE VECTOR INDEX submitted (~ 4.2M rows)
2026-05-21 03:00:01  command returns immediately
2026-05-21 03:11:37  index online — sys.indexes.is_disabled = 0
total cost: 11m 36s wall, ~22 vCore-minutes
```

Smoke test:

KNN-SMOKE.SQL — sql

```sql
DECLARE @q VECTOR(1536) = (
  SELECT embedding FROM search.chunks WHERE chunk_id = 1);

SELECT TOP 5 chunk_id,
       VECTOR_DISTANCE('cosine', embedding, @q) AS dist
FROM search.chunks
ORDER BY VECTOR_DISTANCE('cosine', embedding, @q);
```

Top-1 distance should be `0.0` (the query itself); top-5 returned in < 100ms on a warm cache.

Pitfall: DiskANN at GA is **single-replica** and is dropped + rebuilt on full rebuilds — no online maintenance. Plan a maintenance window for any large delete-heavy rebuild. Also: it's still **public preview** in May 2026; do not rely on its SLA without compensating controls.

08 Re-implement hybrid retrieval with RRF

Goal: a single stored proc that does *full-text + vector KNN + Reciprocal Rank Fusion*, returning a ranked top-K. This is what AI Search gives you for free with `queryType=semantic, vectorQueries=[...]`.

HYBRID_SEARCH.SQL — sql

```sql
CREATE OR ALTER PROCEDURE search.hybrid_search
  @query_text  NVARCHAR(1000),
  @query_vec   VECTOR(1536),
  @top_k       INT = 50,
  @rrf_k       INT = 60        -- RRF smoothing constant; 60 is the canonical default
AS
BEGIN
  ;WITH
  lex AS (
    SELECT TOP (@top_k) c.chunk_id,
           ROW_NUMBER() OVER (ORDER BY KEY_TBL.RANK DESC) AS r
    FROM search.chunks c
    INNER JOIN CONTAINSTABLE(search.chunks, text, @query_text, @top_k) AS KEY_TBL
      ON c.chunk_id = KEY_TBL.[KEY]
  ),
  vec AS (
    SELECT TOP (@top_k) chunk_id,
           ROW_NUMBER() OVER (
             ORDER BY VECTOR_DISTANCE('cosine', embedding, @query_vec)) AS r
    FROM search.chunks
    WHERE embedding IS NOT NULL
  ),
  fused AS (
    SELECT chunk_id, SUM(1.0 / (@rrf_k + r)) AS rrf_score
    FROM (SELECT * FROM lex UNION ALL SELECT * FROM vec) u
    GROUP BY chunk_id
  )
  SELECT TOP (@top_k)
         f.chunk_id, f.rrf_score,
         c.text, d.title, d.source_uri, d.author, d.modified_at
  FROM fused f
  JOIN search.chunks   c ON c.chunk_id = f.chunk_id
  JOIN search.documents d ON d.doc_id = c.doc_id
  ORDER BY f.rrf_score DESC;
END
```

SAMPLE CALL

CALL-HYBRID.SQL **sql**

```sql
DECLARE @qv VECTOR(1536) = /* embed("annual deductible for HMO") via Step 6 */ ...;
EXEC search.hybrid_search
```

```
  @query_text = N'annual deductible HMO',
  @query_vec  = @qv,
  @top_k      = 25;
```

SAMPLE OUTPUT

```
chunk_id  rrf_score   title                              source_uri
--------  ----------  ---------------------------------  ----------------------------
8842      0.03145     Member Benefits 2026               .../mbk-2026.pdf
8841      0.02877     Member Benefits 2026               .../mbk-2026.pdf
14093     0.02560     HMO Plan Summary                   .../hmo-summary.pdf
...
```

Pitfall: RRF requires both ranked lists to use the SAME row identifier. If you join on `doc_id` in one branch and `chunk_id` in the other, fusion is meaningless. Fix scope first.

09 Build a semantic re-ranker — the biggest gap

Goal: rescore the top-50 from Step 8 with a cross-encoder to get AI Search-quality semantic ranking. Highest cost, biggest quality lift.

OPTION A — SELF-HOSTED CROSS-ENCODER (RECOMMENDED FOR COST)

RERANKER_APP.PY — python

```python
from fastapi import FastAPI
from sentence_transformers import CrossEncoder
from pydantic import BaseModel

model = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-12-v2", device="cuda")
app = FastAPI()

class Req(BaseModel):
    query: str
    passages: list[str]

@app.post("/rerank")
def rerank(r: Req):
    scores = model.predict([(r.query, p) for p in r.passages]).tolist()
    order = sorted(range(len(r.passages)), key=lambda i: -scores[i])
    return {"order": order, "scores": [scores[i] for i in order]}
```

DEPLOY.SH

```bash
az containerapp create \
  -g $RG -n reranker \
  --environment search-env \
  --image myacr.azurecr.io/reranker:1.0 \
  --workload-profile-name gpu-nc24 \
  --min-replicas 1 --max-replicas 4 \
  --ingress internal --target-port 8000
```

OPTION B — GPT-4O-MINI SCORER (HIGHER QUALITY, ~20× COST)

GPT_RERANK.PY

```python
prompt = """You are a relevance grader. Given a query and a passage, return
an integer 0..10 for how relevant the passage is. Output only the integer."""

scores = []
for p in passages:
    r = client.chat.completions.create(model="gpt-4o-mini",
        messages=[{"role":"system","content":prompt},
                  {"role":"user","content":f"QUERY: {query}\nPASSAGE: {p}"}],
        max_tokens=2, temperature=0)
    scores.append(int(r.choices[0].message.content.strip()))
```

WIRE INTO THE APP TIER

SEARCH_API.PY

```python
def search(query: str, top_k: int = 10):
    qv = embed(query)
    candidates = sql.exec("EXEC search.hybrid_search ?, ?, 50", query, qv)
    rerank = httpx.post("https://reranker/rerank",
        json={"query": query, "passages": [c.text for c in candidates]}).json()
    return [candidates[i] for i in rerank["order"][:top_k]]
```

Pitfall: running the cross-encoder on the FULL result set instead of top-50 destroys latency. The whole point is "cheap recall (hybrid) + expensive precision (reranker on 50 candidates)."

10 Recreate scoring profiles and synonyms

Goal: field-weighting, freshness boosts, tag boosts, and synonym expansion — features AI Search exposes as JSON config.

SCORING EXPRESSION IN T-SQL

SCORING-PROFILE.SQL — sql

```sql
-- Compose a final score from RRF + freshness boost + tag boost
SELECT TOP 25
  f.chunk_id, f.rrf_score,
  freshness_boost = CAST(
    1.0 / (1 + DATEDIFF(DAY, d.modified_at, SYSUTCDATETIME()) / 30.0) AS FLOAT),
  tag_boost = CASE
    WHEN JSON_VALUE(d.tags, '$[0]') IN ('benefits','formulary') THEN 1.3
    ELSE 1.0 END,
  final_score = f.rrf_score
              * (1 + 0.4 * (1.0 / (1 + DATEDIFF(DAY, d.modified_at, SYSUTCDATETIME()) / 30.0)))
              * (CASE WHEN JSON_VALUE(d.tags,'$[0]') IN ('benefits','formulary')
                      THEN 1.3 ELSE 1.0 END)
FROM fused f
JOIN search.documents d ON ...
ORDER BY final_score DESC;
```

SYNONYMS — FTS THESAURUS

TSENU.XML — xml

```xml
<XML ID="Microsoft Search Thesaurus">
  <thesaurus xmlns="x-schema:tsSchema.xml">
    <diacritics_sensitive>0</diacritics_sensitive>
    <expansion>
      <sub>HMO</sub><sub>Health Maintenance Organization</sub>
    </expansion>
    <expansion>
      <sub>PCP</sub><sub>primary care provider</sub><sub>primary care physician</sub>
    </expansion>
  </thesaurus>
</XML>
```

LOAD-THESAURUS.SQL — sql

```
EXEC sys.sp_fulltext_load_thesaurus_file 1033, @LoadOnly = 0;
-- Re-issue queries; FORMSOF(THESAURUS, 'HMO') now also matches 'Health Maintenance Organization'
```

11 Recreate facets, suggesters, and autocomplete

Goal: the "filter pills" + autocomplete dropdown UX. AI Search ships these as first-class endpoints; you build them.

FACETS

FACETS.SQL — sql

```sql
-- Counts by author and language, scoped to a result set CTE
WITH hits AS (
  SELECT d.doc_id, d.author, d.language
  FROM search.documents d JOIN search.chunks c ON c.doc_id = d.doc_id
  WHERE c.chunk_id IN (/* result chunk_ids from Step 8 */)
)
SELECT 'author' AS facet, author AS value, COUNT(*) AS cnt
FROM hits GROUP BY author
UNION ALL
SELECT 'language', language, COUNT(*) FROM hits GROUP BY language
ORDER BY facet, cnt DESC;
```

SUGGESTER / AUTOCOMPLETE

SUGGESTER.SQL — sql

```sql
-- Materialized view of distinct title prefixes
CREATE TABLE search.title_prefixes (prefix NVARCHAR(64) PRIMARY KEY);
INSERT INTO search.title_prefixes
SELECT DISTINCT LOWER(LEFT(title, n))
FROM search.documents
CROSS APPLY (VALUES (3),(5),(8),(12),(20)) AS p(n)
WHERE title IS NOT NULL;

-- Autocomplete query
SELECT TOP 10 prefix
FROM search.title_prefixes
```

```
WHERE prefix LIKE @typed + '%'
ORDER BY prefix;
```

Pitfall: querying facets against the full `chunks` table for every search request is slow. Always scope to the candidate set returned by Step 8.

12 Wire up observability & query telemetry

Goal: per-query latency, candidate counts, click-through, and a relevance dashboard. AI Search Insights is gone — you stitch it together.

ENABLE PLATFORM DIAGNOSTICS

DIAG.SH — bash

```bash
LAW=law-search-prod
az monitor log-analytics workspace create -g $RG -n $LAW

az monitor diagnostic-settings create \
  --resource $(az sql db show -g $RG -s $SRV -n $DB --query id -o tsv) \
  --name diag-all \
  --workspace $LAW \
  --logs '[{"category":"QueryStoreRuntimeStatistics","enabled":true},
           {"category":"AutomaticTuning","enabled":true},
           {"category":"Errors","enabled":true}]' \
  --metrics '[{"category":"Basic","enabled":true}]'
```

APP-TIER QUERY LOG TABLE

TELEMETRY.SQL — sql

```sql
CREATE TABLE search.query_log (
  query_id      UNIQUEIDENTIFIER PRIMARY KEY DEFAULT NEWID(),
  ts            DATETIME2(3) NOT NULL DEFAULT SYSUTCDATETIME(),
  user_id       NVARCHAR(128),
  query_text    NVARCHAR(1000),
  candidates    INT,
  top_chunk_id  BIGINT,
  rrf_top       FLOAT,
  rerank_top    FLOAT,
  total_ms      INT,
  embed_ms      INT,
```

```
  hybrid_ms      INT,
  rerank_ms      INT,
  clicked_at     DATETIME2(3) NULL,
  clicked_rank   INT NULL
);
```

RELEVANCE DASHBOARD (KQL ON LOG ANALYTICS)

RELEVANCE.KQL — sql

```
// p50/p95 latency by hour
AppRequests
| where Name == "POST /search"
| summarize p50=percentile(DurationMs,50), p95=percentile(DurationMs,95) by bin(TimeGenerated,1h)
| render timechart

// CTR at rank 1
customEvents
| where name == "search_click"
| summarize ctr = todouble(countif(toint(customDimensions.clicked_rank)==1))/count()
            by bin(timestamp,1d)
```

13 Security, network, and identity

Goal: the controls a regulated workload (HIPAA / HITRUST / FedRAMP) needs — TDE+CMK, private endpoint, Entra-only, RLS, column-level encryption.

TDE WITH CUSTOMER-MANAGED KEY

CMK.SH — bash

```
KV=kv-search-prod
KEY=cmk-search-2026

az keyvault create -g $RG -n $KV -l $LOC --enable-purge-protection true
az keyvault key create --vault-name $KV -n $KEY --kty RSA --size 3072

# Grant the SQL server's managed identity wrap/unwrap/get
az keyvault set-policy -n $KV \
  --object-id $(az sql server show -g $RG -n $SRV --query identity.principalId -o tsv) \
  --key-permissions wrapKey unwrapKey get
```

```
KEY_URI=$(az keyvault key show --vault-name $KV -n $KEY --query key.kid -o tsv)

az sql server tde-key set -g $RG -s $SRV --server-key-type AzureKeyVault \
  --kid $KEY_URI

az sql db tde set -g $RG -s $SRV -n $DB --status Enabled
```

PRIVATE ENDPOINT

PE.SH **bash**

```
az network private-endpoint create -g $RG -n pe-sql-search \
  --vnet-name vnet-search --subnet snet-data \
  --private-connection-resource-id $(az sql server show -g $RG -n $SRV --query id -o tsv) \
  --group-id sqlServer \
  --connection-name pe-sql-search-conn

az sql server update -g $RG -n $SRV --enable-public-network false
```

ROW-LEVEL SECURITY (MULTI-TENANT PATTERN)

RLS.SQL **sql**

```
CREATE SCHEMA sec;
GO
CREATE FUNCTION sec.fn_tenant_predicate(@tenant NVARCHAR(64))
RETURNS TABLE WITH SCHEMABINDING AS
RETURN SELECT 1 AS ok
WHERE @tenant = CAST(SESSION_CONTEXT(N'tenant_id') AS NVARCHAR(64));
GO

ALTER TABLE search.documents ADD tenant_id NVARCHAR(64) NOT NULL DEFAULT 'public';

CREATE SECURITY POLICY sec.documents_filter
ADD FILTER PREDICATE sec.fn_tenant_predicate(tenant_id) ON search.documents,
ADD BLOCK  PREDICATE sec.fn_tenant_predicate(tenant_id) ON search.documents
WITH (STATE = ON);
```

APP SETS THE TENANT BEFORE EVERY QUERY

APP.PY **python**

```
cur.execute("EXEC sp_set_session_context @key=N'tenant_id', @value=?", tenant_id)
# All subsequent queries on search.documents are filtered automatically
```

14 If you used agentic retrieval — rebuild it

Goal: multi-turn query planning + reasoning + retrieval. AI Search calls this "agentic retrieval"; you rebuild it in your app tier with an LLM orchestrator.

AGENTIC.PY — python

```python
from semantic_kernel import Kernel
from semantic_kernel.connectors.ai.open_ai import AzureChatCompletion
from semantic_kernel.functions import kernel_function

kernel = Kernel()
kernel.add_service(AzureChatCompletion(deployment_name="gpt-4o",
    endpoint=os.environ["AOAI_ENDPOINT"], ad_token_provider=token))

class SearchPlugin:
    @kernel_function(description="Hybrid + rerank search against SQL")
    def search(self, query: str, top_k: int = 10) -> str:
        return json.dumps(do_hybrid_search(query, top_k))  # wraps Step 9

kernel.add_plugin(SearchPlugin(), "kb")

PLANNER = """You are a healthcare claims expert. Decompose the user's question
into 1-3 search sub-queries against the knowledge base, run them via the
kb.search function, then synthesize a grounded answer with citations."""

result = await kernel.invoke_prompt(
    prompt=PLANNER + "\nUser: " + user_q,
    arguments=KernelArguments(settings={"function_choice_behavior": "auto"}))
```

SAMPLE RUN

```
User: "Am I covered for outpatient PT and what's the copay for an HMO member?"

Planner -> 2 sub-queries:
  1. "outpatient physical therapy coverage HMO"
  2. "HMO copay outpatient therapy"
```

```
Tool calls:
  kb.search("outpatient physical therapy coverage HMO", top_k=5)
  kb.search("HMO copay outpatient therapy", top_k=5)

Final answer (grounded, with [doc:mbk-2026.pdf §4.3] citations) returned in 4.1s
```

Pitfall: let the planner call `kb.search` at most 3 times per turn. Without a hard cap, GPT will fan out to 10+ sub-queries and blow your retrieval cost budget.

15 Cut over with shadow traffic

Goal: prove the SQL-based system meets or beats AI Search on a representative query log *before* turning users over. NDCG@10 and MRR are the standard metrics.

SHADOW HARNESS

SHADOW.PY — python

```python
import math, json, statistics
from azure.search.documents import SearchClient
from azure.identity import DefaultAzureCredential

ais = SearchClient(endpoint=AIS_ENDPOINT, index_name="prod-index",
                   credential=DefaultAzureCredential())

def ndcg(judged, returned, k=10):
    dcg = sum(judged.get(d, 0) / math.log2(i + 2) for i, d in enumerate(returned[:k]))
    ideal = sorted(judged.values(), reverse=True)[:k]
    idcg = sum(g / math.log2(i + 2) for i, g in enumerate(ideal))
    return dcg / idcg if idcg else 0.0

with open("query_judgments.jsonl") as f:
    rows = [json.loads(l) for l in f]

ais_ndcg, sql_ndcg = [], []
for r in rows:
    q, judged = r["query"], r["judgments"]   # {doc_id: rel (0-3)}
    ais_top = [d["doc_id"] for d in ais.search(q, top=10)]
    sql_top = [d["doc_id"] for d in do_hybrid_search(q, top_k=10)]
    ais_ndcg.append(ndcg(judged, ais_top))
    sql_ndcg.append(ndcg(judged, sql_top))
```

```
print(f"AI Search  NDCG@10: {statistics.mean(ais_ndcg):.4f}")
print(f"SQL Hybrid NDCG@10: {statistics.mean(sql_ndcg):.4f}")
print(f"Delta: {(statistics.mean(sql_ndcg) - statistics.mean(ais_ndcg)) * 100:+.2f} pts")
```

SAMPLE EXECUTION

```
$ python shadow.py
Evaluating 1,200 queries...
AI Search  NDCG@10: 0.7421
SQL Hybrid NDCG@10: 0.7388
Delta: -0.33 pts

Verdict: within 0.5 pts -> GO for cutover
```

CUTOVER PLAN

1. **Day -14:** deploy SQL stack to prod, dual-write ingestion (writes go to BOTH AI Search and SQL).
2. **Day -7:** dual-read with shadow eval (above) running on every real query in the background; log deltas.
3. **Day 0:** flip read traffic to SQL via feature flag (1% → 10% → 50% → 100% over 48h).
4. **Day +7:** stop dual-write; decommission the AI Search service.

> **Go / no-go gate:** SQL NDCG@10 within 0.5 pts of AI Search AND p95 latency < AI Search p95 + 100ms. Anything worse, do not cut over — invest more in the re-ranker (Step 9).

Azure AI Search — per Search Unit (East US, May 2026)

TIER	$/SU/HR	$/SU/MO (730 HR)	STORAGE / PARTITION	MAX PARTITIONS	MAX SUS / SERVICE
Basic	$0.101	$73.73	15 GB	3	9
Standard S1	$0.336	$245.28	160 GB	12	36
Standard S2	$1.344	$981.12	512 GB	12	36
Standard S3	$2.688	$1,962.24	1 TB	12	36
Storage Opt L1	$3.839	$2,802.47	2 TB	12	36
Storage Opt L2	$7.677	$5,604.21	4 TB	12	36

SU = partitions × replicas. Service-wide cap is 36 SUs (so with 4 replicas, the maximum partition count is 9 — bumping partitions further requires upgrading the tier). **Azure AI Search does not offer reserved capacity / 1-yr or 3-yr commitment pricing** — all billing is hourly PAYG.

Vector index size cap per partition (separate from data storage cap): Basic 5 GB, S1 35 GB, S2 150 GB, S3 / L2 300 GB, L1 150 GB.

Add-ons

METER	PRICE
Semantic ranker (Standard tier)	$1.00 / 1 000 queries
Semantic ranker (Free tier)	First 1 000 queries / mo free
Document cracking — image extraction	$1.00 / 1 000 (volume discounts apply)
Agentic retrieval — medium reasoning	$0.0001 / 1 000 tokens

Azure SQL Database Hyperscale (East US, May 2026)

COMPUTE SERIES	$/VCORE/HR (PAYG)	1-YR RESERVED	3-YR RESERVED

Standard-series Gen5	$0.18266	~$0.1187	~$0.0822
Premium-series	$0.18266	~$0.1187	~$0.0822
Premium-series Memory-Optimized	$0.25572	~$0.1663	~$0.1150
DC-series (confidential)	$0.36500	$0.2374	$0.1644

Each HA replica and named replica is billed at the same compute rate as the primary. Reserved capacity discounts are approximate (~35% for 1-yr, ~55% for 3-yr).

Storage & backup

METER	PRICE
Hyperscale data storage (Single DB)	$0.10 / GB / month
Hyperscale data storage (Elastic Pool)	$0.25 / GB / month
Hyperscale I/O	$0.20 / 1 M I/O ops
Backup storage (LRS / ZRS short-term)	Included up to 100% of DB size
LTR backup (LRS / ZRS)	$0.025 / GB / month
LTR backup (RA-GRS / RA-GZRS)	$0.05 / GB / month

Source: Azure Retail Prices API (`prices.azure.com/api/retail/prices`), East US region, retrieved May 20, 2026.